
GENERAL RELATIVITY

Tutorial problem set 15, 15.01.2021, with solutions.

■ **PROBLEM 37** Dimming of objects falling into a black hole.

Consider an observer outside of a Schwarzschild black hole sitting at constant Schwarzschild coordinates $(r_0, \vartheta = \frac{\pi}{2}, \varphi = 0)$. At some point $t = 0$ he releases a beacon that starts falling towards the black hole, while constantly emitting a monochromatic light signal (of frequency ν_{em}) back to the observer (in the radial direction). In this problem you will determine what is the signal received by the observer.

- (a) One generally needs to be careful what kind of velocities one defines in GR. Here we can define at least three ones for the beacon of certain use:
- (i) The coordinate velocity, $v_r = \frac{dr}{dt}$, that measures what is the coordinate distance dr that the beacon travels during coordinate time dt ,
 - (ii) The r -component of the 4-velocity, $u_r = \frac{dr}{d\tau}$, that measures the coordinate distance dr the beacon travels during its own proper time interval $d\tau$,
 - (iii) The 'velocity of a stationary observer', $w_r = \frac{dr}{d\tau_R}$, which is the velocity of the beacon that would be measured by an observer at the fixed coordinate R , at the moment the beacon is passing him by, where τ_R is the proper time of that observer.

Calculate these three velocities as functions of r . Rather than solving the geodesic equations, make use of the Schwarzschild line element, and the conserved quantity E derived in problem 32 of the tutorial problem set 12. Fix E by imposing initial conditions.

- (b) Find the relation between the coordinate time and the proper time of a static observer at R , and between coordinate time and the proper time of the beacon at r .

In order to find the frequency of the signal received by the observer at r_0 , we will employ a similar strategy as in problem 33 of the tutorial problem set 12. We will relate the proper time (of the beacon) between the emission of the two successive light crests, and the proper time of the observer at r_0 between the absorption of these two crests.

- (c) The beacon emits the first crest at (t_{em1}, r_{em1}) , and the second one (t_{em2}, r_{em2}) . The coordinate time between the emission of two successive crests is $\Delta t_{em} = t_{em2} - t_{em1}$, and during that time the beacon has moved by $\Delta r_{em} = r_{em2} - r_{em1}$. Calculate the coordinate time Δt_{re} between the reception of the two crests by the observer at r_0 . In order to do that, calculate the coordinate time it takes each crest to reach the observer.
- (d) We can safely assume that the time interval between the emission of the two crests is much smaller than any other relevant scale. Expand the expression obtained in (c) to linear order in Δt_{em} , and express the ratio $\Delta t_{re}/\Delta t_{em}$ in terms of r_o , $r_S = 2G_N M$ and $r_{em1} \equiv r$.

- (e) Using the expressions found in (b) and (d) show that the frequency received by the observer is

$$\nu_{re} = \nu_{em} \sqrt{\frac{A(r)}{A(r_o)}} \times \frac{\sqrt{A(r)}}{\sqrt{A(r_o)} + \sqrt{A(r_o) - A(r)}} , \quad (37.1)$$

where $A(r) = 1 - \frac{r_S}{r}$ and ν_{em} and ν_{re} denote the emitted and observed frequency, respectively. Compare this result to the one obtained in problem 33 of the tutorial problem set 12. Interpret where does the extra term here come from.

SOLUTION: (a) From problem 32 we know that the conserved energy and angular momentum (at $\vartheta = \pi/2$) are,

$$E = A(r) \frac{dt}{d\tau}, \quad L = r^2 \frac{d\varphi}{d\tau}, \quad A(r) = 1 - \frac{r_S}{r}, \quad r_S = 2G_N M. \quad (37.2)$$

Since the beacon is dropped radially towards the black hole, its angular momentum is $L = 0$. The line element in Schwarzschild coordinates is then,

$$-d\tau^2 = -A(r)dt^2 + \frac{dr^2}{A(r)}, \quad (37.3)$$

from where we obtain the radial 4-velocity,

$$u_r = \frac{dr}{d\tau} = -\sqrt{E^2 - A(r)}. \quad (37.4)$$

The coordinate radial velocity is then,

$$v_r = \frac{dr}{dt} = \frac{dr}{d\tau} \frac{d\tau}{dt} = -A(r) \sqrt{1 - \frac{A}{E^2}}. \quad (37.5)$$

Finally, the coordinate velocity as seen from the point of view of the observer at rest can be obtained from the observation that for that observer (for which $dr = 0$),

$$-d\tau^2 = -A(r)dt_R^2 \quad (37.6)$$

such that

$$w_r = \frac{dr}{dt_R} = \frac{dr}{d\tau} \frac{d\tau}{dt_R} = -\sqrt{E^2 - A(r)} \sqrt{A(r)} = -\sqrt{A(r)[E^2 - A(r)]}. \quad (37.7)$$

- (b) The first part of the question was indicated in (37.6), therefore

$$dt_R = \frac{d\tau}{\sqrt{A(r)}} = \frac{dt}{\sqrt{A(r)}} \frac{d\tau}{dt} = \frac{dt}{\sqrt{A(r)}} \frac{A(r)}{E} = \frac{\sqrt{A(r)}}{E} dt \implies t_R = \frac{\sqrt{A(r)}}{E} (t + \text{const.}) . \quad (37.8)$$

Next, beacon's time is proper time $dt_B = d\tau$, and therefore we have,

$$dt_B = d\tau = \frac{d\tau}{dt} dt = \frac{A(r)}{E} dt \implies t_B = \frac{A(r)}{E} (t + \text{const.}) . \quad (37.9)$$

(c) When measured in Schwarzschild coordinates, the light moves on geodesics given by,

$$0 = -A(r)dt^2 + \frac{dr^2}{A(r)} \implies \frac{dr}{A(r)} = \pm dt. \quad (37.10)$$

When integrated this gives (the sign below is chosen such that the signal is outgoing),

$$r(t) + r_S \ln \left(\frac{r(t)}{r_S} - 1 \right) = t + \text{const.}, \quad (37.11)$$

from where we conclude that the coordinate times at which the observer at r_0 receives the signals are,

$$r_o - r_{em1} + r_S \ln \left(\frac{r_0 - r_S}{r_{em1} - r_S} \right) = t_{re1} - t_{em1}, \quad r_o - r_{em2} + r_S \ln \left(\frac{r_0 - r_S}{r_{em2} - r_S} \right) = t_{re2} - t_{em2}. \quad (37.12)$$

Upon subtracting the two one obtains,

$$-\Delta r_{em} + r_S \ln \left(\frac{r_{em1} - r_S}{r_{em2} - r_S} \right) = \Delta t_{re} - \Delta t_{em} \implies \Delta t_{re} = \Delta t_{em} - \Delta r_{em} + r_S \ln \left(\frac{r_{em1} - r_S}{r_{em2} - r_S} \right), \quad (37.13)$$

where $\Delta t_{re} = t_{re2} - t_{re1}$. From (37.13) we see that $\Delta t_{re} > \Delta t_{em}$ (because $\Delta r_{em} < 0$), implying that the intervals Δt are increasing as the position r decreases. Another way of expressing this result would be in terms of the proper time of the beacon, $\Delta t_B = \Delta \tau$ given in (37.9). We leave it as an exercise to the reader.

(d) If Δr_{em} is small ($\Delta r_{em} \ll r_{em1} - r_S$), Eq. (37.13) can be approximated by,

$$\Delta t_{re} \approx \Delta t_{em} - \Delta r_{em} \left[\frac{r_{em1}}{r_{em1} - r_S} \right] \implies \frac{\Delta t_o}{\Delta t_{em}} \approx 1 - \frac{\Delta r_{em}}{\Delta t_{em}} \left[\frac{r_{em1}}{r_{em1} - r_S} \right] = 1 - \frac{1}{A(r_{em1})} \frac{\Delta r_{em}}{\Delta t_{em}}, \quad (37.14)$$

where

$$\frac{\Delta r_{em}}{\Delta t_{em}} \approx \frac{dr}{dt} = -\frac{A(r)}{E} \sqrt{E^2 - A(r)}, \quad (37.15)$$

where E is the energy of the beacon (per unit mass). Assuming that the beacon has a vanishing velocity at $r = r_o$, we also have that, $E = \sqrt{A(r_o)} = \sqrt{1 - \frac{r_S}{r_o}} < 1$. Inserting (37.15) into (37.14) one obtains,

$$\frac{\Delta t_{re}}{\Delta t_{em}} \approx 1 + \sqrt{1 - \frac{A(r)}{E^2}} = 1 + \sqrt{1 - \frac{A(r)}{A(r_o)}} > 1, \quad (37.16)$$

such that the observer at $r = r_o$ receives signals in intervals exhibiting a time dilatation.

(e) First note that the phase between two crests Φ is an invariant, and thus $\Phi = A(r_{em})\nu_{em}\Delta t_{em} = A(r_o)\nu_{re}\Delta t_{re}$, such that, $\nu_{re}/\nu_{em} = (A(r_{em})/A(r_o))(\Delta t_{em}/\Delta t_{re})$. Inserting this into (37.16) yields,

$$\frac{\nu_{re}}{\nu_{em}} = \frac{A(r)}{A(r_o)} \frac{\Delta t_{em}}{\Delta t_{re}} = \frac{A(r)}{A(r_o)} \frac{\sqrt{A(r_o)}}{\sqrt{A(r_o)} + \sqrt{A(r_o) - A(r)}}, \quad (37.17)$$

which agrees with (37.1). The result (37.17) differs from the one obtained in problem 13 because that result was obtained for a source at a constant distance from the black hole, which is to be contrasted with the moving beacon of this problem, which is freely falling.

■ **PROBLEM 38** Motion of a particle in Kerr space-time.

The line element of the Kerr space-time in Boyer-Lindquist coordinates is,

$$ds^2 = - \left[1 - \frac{2G_N M r}{\rho^2} \right] dt^2 - \frac{4G_N M a r \sin^2(\theta)}{\rho^2} dt d\phi + \frac{\rho^2}{\Delta} dr^2 + \rho^2 d\theta^2 \\ + \left[r^2 + a^2 + \frac{2G_N M r a^2 \sin^2(\theta)}{\rho^2} \right] \sin^2(\theta) d\phi^2 , \quad (38.1)$$

where

$$\rho^2 = r^2 + a^2 \cos^2(\theta) , \quad \Delta = r^2 - 2G_N M r + a^2 . \quad (38.2)$$

In this problem you will examine the orbits of massive particles and light particles that lie in the equatorial plane $\theta = \frac{\pi}{2}$.

- (a) Write the metric (38.1) in the equatorial plane $\theta = \frac{\pi}{2}$.
 (b) Since the metric (38.1) does not depend on coordinates t and ϕ we can immediately conclude that there are two Killing vectors in this space-time,

$$K^\mu = (1, 0, 0, 0) , \quad R^\mu = (0, 0, 0, 1) , \quad (38.3)$$

associated with the time translation symmetry and the axial rotational symmetry, respectively. When there are Killing vectors, there are conserved quantities along geodesics associated to them. In this case those are energy and the z -component of the angular momentum,

$$E = -K_\mu \frac{dx^\mu}{d\lambda} , \quad \ell = R_\mu \frac{dx^\mu}{d\lambda} . \quad (38.4)$$

Express $dt/d\lambda$ and $d\phi/d\lambda$ in terms of E and ℓ and the radial coordinate r .

- (c) Using the results from (a) and (b) find the effective potential $W_{\text{eff}}(E, \ell; r)$ for the motion of a massless particle in the equatorial plane,

$$\frac{1}{2} \left(\frac{dr}{d\lambda} \right)^2 + W_{\text{eff}}(E, \ell; r) = \frac{E^2}{2} , \quad (38.5)$$

and show that it can be written as,

$$W_{\text{eff}} = \frac{\ell - aE}{2} \left[\frac{\ell + aE}{r^2} - \frac{r_S(\ell - aE)}{r^3} \right] , \quad (38.6)$$

such that $W_{\text{eff}} = 0$ for $\ell = aE$. What is the radius of circular orbits of light? Are they stable or unstable? Show that there exists a minimum angular momentum ℓ for which such an orbit exists. Calculate that ℓ and express it in terms of E , a , ℓ and r_S .

Notice that for circular orbits E is not an independent quantity. Find $E = E(r_S, \ell)$ for a maximally spinning black hole, $a = r_S/2$, and determine the radius of the circular orbit r_0 for $\ell > 0$ and for $\ell < 0$.

- (d) Find the effective potential for massive particles (take $\lambda = \tau$, where τ denotes proper time),

$$\frac{1}{2} \left(\frac{dr}{d\tau} \right)^2 + V_{\text{eff}}(E, \ell; r) = \frac{E^2 - 1}{2} , \quad (38.7)$$

and show that it can be written as,

$$V_{\text{eff}} = -\frac{r_S}{2r} + \frac{1}{2} (\ell^2 - a^2(E^2 - 1)) \frac{1}{r^2} - \frac{1}{2} (\ell - aE\ell)^2 \frac{r_S}{r^3} . \quad (38.8)$$

- (e) Consider a massive particle initially infalling radially ($\ell = 0$) towards the Kerr black hole, and with negligible initial kinetic energy ($E = 1$ – why this value?). Kerr space-time is stationary, but it is not static (black hole is rotating), and as an effect it will drag the initially radially infalling particle with its rotation, i.e. $d\phi/dr$ will not be constant. Show that

$$\frac{d\phi}{dr} = -\frac{2MG_N a}{r\Delta} \left[\frac{2MG_N}{r} \left(1 + \frac{a^2}{r^2} \right) \right]^{-1/2}. \quad (38.9)$$

The angle $\Delta\phi(r_0)$ the trajectory of the particle has swept from $r = \infty$ to some $r = r_0$ can in principle be calculated by integrating this expression from ∞ to r_0 . Will this particle eventually fall into the black hole?

- (f) Make use of (38.8) to calculate the stable and unstable circular orbits for massive particles and show that they are given by,

$$r_{\pm} = \frac{1}{r_S} [\ell^2 - a^2(E^2 - 1)] \pm \sqrt{\frac{1}{r_S^2} [\ell^2 - a^2(E^2 - 1)]^2 - 3(\ell - aE)^2}. \quad (38.10)$$

Are these orbits stable or unstable? What are their angular velocities?

- (g) Consider an extremal Kerr black hole for which $a = G_N M$. Make use of (38.10) to find the circular orbits and show that, in the limit when $r_c = \frac{2\ell^2}{r_S} \gg r_S$ and $E \rightarrow 1$ they reduce to,

$$r_+ \simeq r_c \left[1 - \frac{3r_S}{2r_c} + \frac{3}{2} \left(\frac{2r_S}{r_c} \right)^{\frac{3}{2}} \right], \quad r_- \approx \frac{3}{2} r_S \left[1 - \sqrt{\frac{2r_S}{r_c}} + \frac{2r_S}{r_c} \right]. \quad (38.11)$$

Compare these orbits with those of a Schwarzschild black hole and identify the differences.

SOLUTION: (a) In the equatorial plane $\theta = \pi/2$, the metric (38.1) simplifies to,

$$ds^2 = - \left[1 - \frac{2G_N M}{r} \right] dt^2 - \frac{4G_N M a}{r} dt d\phi + \frac{r^2}{\Delta} dr^2 + \left[r^2 + a^2 + \frac{2G_N M a^2}{r} \right] d\phi^2, \quad (38.12)$$

where

$$\rho = r, \quad \Delta = r^2 - 2G_N M r + a^2. \quad (38.13)$$

- (b) From (38.3) and (38.4) we infer,

$$E = -K_t \frac{dt}{d\lambda} - K_\phi \frac{d\phi}{d\lambda} = \left(1 - \frac{r_S}{r} \right) \dot{t} + \frac{r_S a}{r} \dot{\phi}, \quad (38.14)$$

$$\ell = R_t \frac{dt}{d\lambda} + R_\phi \frac{d\phi}{d\lambda} = -\frac{r_S a}{r} \dot{t} + \left(r^2 + a^2 + \frac{r_S a^2}{r} \right) \dot{\phi}, \quad (38.15)$$

where $r_S = 2G_N M$, $\dot{t} = dt/d\lambda$, $\dot{\phi} = d\phi/d\lambda$. Eqs. (38.14–38.15) can be inverted to give,

$$\dot{t} = \frac{\left(r^2 + a^2 + \frac{r_S a^2}{r} \right) E - \frac{r_S a}{r} \ell}{\Delta}, \quad (38.16)$$

$$\dot{\phi} = \frac{\frac{r_S a}{r} E + \left(1 - \frac{r_S}{r} \right) \ell}{\Delta}. \quad (38.17)$$

(c) Upon inserting (38.16–38.17) into the line element (38.12) one obtains,

$$\epsilon = \left(1 - \frac{r_S}{r}\right) \dot{t}^2 + \frac{2r_S a}{r} \dot{t} \dot{\phi} - \frac{r^2}{\Delta} \dot{r}^2 - \left(r^2 + a^2 + \frac{r_S a^2}{r}\right) \dot{\phi}^2, \quad (38.18)$$

where $\epsilon = 1$ ($\epsilon = 0$) for massive (massless) particles. Inserting Eqs. (38.16–38.17) into (38.18) and multiplying by $-\frac{\Delta}{2r^2}$ one obtains,

$$W_{\text{eff}} = \frac{E^2}{2} + \frac{\Delta}{2r^2} \left\{ \epsilon + \frac{-\left(r^2 + a^2 + \frac{r_S a^2}{r}\right) E^2 + 2\frac{r_S}{r} a E \ell + \left(1 - \frac{r_S}{r}\right) \ell^2}{\Delta} \right\} \quad (38.19)$$

The effective potential for light is obtained by setting $\epsilon \rightarrow 0$,

$$W_{\text{eff}} = \frac{-a^2 \left(1 + \frac{r_S}{r}\right) E^2 + 2\frac{r_S}{r} a E \ell + \left(1 - \frac{r_S}{r}\right) \ell^2}{2r^2} = \frac{\ell^2 - a^2 E^2}{2r^2} - \frac{r_S (\ell - aE)^2}{2r^3}. \quad (38.20)$$

From here we see that $W'_{\text{eff}}(r) = 0$ implies,

$$r_0 = \frac{3}{2} r_S \frac{\ell - aE}{\ell + aE}, \quad (38.21)$$

is the radius of the circular light orbit in the equatorial plane.

To answer the second part of the question, note that $r_0 > r_+^H = \frac{r_S}{2} [1 + \sqrt{1 - (2a/r_S)^2}]$, from which it follows,

$$\frac{\ell}{E} > a \frac{4 + \sqrt{1 - \left(\frac{2a}{r_S}\right)^2}}{2 - \sqrt{1 - \left(\frac{2a}{r_S}\right)^2}}, \quad (38.22)$$

which for maximally charged black hole reduces to, $\ell/E > r_S$, or $E < \ell/r_S \implies E^2 < r_c/(2r_S)$, where $r_c = 2\ell^2/r_S$ is the classical circular orbit radius. On the other hand, in order for an orbit to be circular, $\dot{r} = 0$, or equivalently $W_{\text{eff}}(r_0) = E^2/2$. Upon insertnig (38.21) into this yields the following condition on energy,

$$W_{\text{eff}}(r_0) = \frac{2}{27} \frac{(\ell + aE)^3}{r_S^2 (\ell - aE)} = \frac{E^2}{2}. \quad (38.23)$$

Furthermore, the positivity of W_{eff} tells us that we must seek an energy in the interval, $-\frac{\ell}{a} < E < \frac{\ell}{a}$. Eq. (38.23) can be recast as a cubic equation for $x = aE/\ell$,

$$(1+x)^3 - y^2 x^2 (1-x) = 0, \quad y^2 = \frac{27}{4} \frac{r_S^2}{a^2} \implies y = \sqrt{3} \frac{3r_S}{2a}. \quad (38.24)$$

This equation is simple to solve for the case of a maximally spinning black hole, in which case $a = r_S/2$ and $r_{\pm}^H = r_S/2$, such that $y = 3\sqrt{3}$ and Eq. (38.24) simplifies to,

$$(1+x)^3 - 27x^2(1-x) = 0 \Leftrightarrow (2x-1)^2(7x+1) = 0, \quad y = 3\sqrt{3}, \quad x = \frac{r_S E}{2\ell}. \quad (38.25)$$

The double postive root gives, $E = \ell/r_S = \sqrt{r_c}/(\sqrt{2}r_S)$, for which $r_0 = r_S/2 = r_{\pm}^H$, which states that photons can circle around a maximally charged black hole at at the event horizon, $r = r_{\pm}^H$ (that that is true can be also follows from the observation that both light cones are vertical at

$r = r_{\pm}^H$, *i.e.* $dt/dr = \infty$). The third root $x = -1/7$ is also interesting as the corresponding energy is negative, $E = -2\ell/(7r_S)$, $r_0 = 2r_S = 4r_{\pm}^H > r_{\pm}^H$, and $W_{\text{eff}} = \frac{2\ell^2}{49r_S^2} = \frac{r_c}{49r_S}$. Since the energy cannot be negative, we conclude that this solution corresponds to $\ell < 0$, $E = 2|\ell|/(7r_S)$. This means that there are two distinct cases: (1) photons corotate with the black hole, in which case the radius of the unstable photon orbit coincides with the external event horizon, $r_0 = r_S/2$, and (2) photons rotate in the opposite sense from the black hole, in which case the unstable orbit is larger, $r_0 = 2r_S$.

We leave as an exercise to the reader to discuss the more general case of non-maximally charged black holes, $a < r_S/2$. While it is hard to study this case in general, the two cases are rather simple, namely $2a/r_S \ll 1$ and $0 < 1 - \frac{2a}{r_S} \ll 1$.

(d) For massive particles $\epsilon = 1$ and we then have from (38.19),

$$V_{\text{eff}} = \frac{E^2 - 1}{2} + \frac{\Delta}{2r^2} + \frac{-\left(r^2 + a^2 + \frac{r_S}{r}a^2\right)E^2 + 2\frac{r_S}{r}aE\ell + \left(1 - \frac{r_S}{r}\right)\ell^2}{2r^2}. \quad (38.26)$$

(e) At asymptotically large distances, $E = \gamma$, $\vec{u}^2 = \|\gamma\vec{v}\|^2 = \gamma^2 - 1$, such that $u^2 = -E^2 + \|\vec{u}\|^2 = -\gamma^2 + (\gamma^2 - 1) = -1$, as it should be, where $\gamma = \sqrt{1 - \|\vec{v}\|^2}$ and $\vec{v}$ are the Lorentz gamma particle's velocity at spatial infinity. Clearly, for particle with a vanishing initial velocity, $\vec{v} = 0$, $\gamma = E = 1$. For a particle with an initial angular momentum $\ell = 0$ and $E = 1$, we have

$$\frac{\dot{r}^2}{2} + V_{\text{eff}} = 0, \quad V_{\text{eff}} = \frac{\Delta}{2r^2} \left\{ 1 + \frac{-\left(r^2 + a^2 + \frac{r_S}{r}a^2\right)}{\Delta} \right\} = -\frac{\frac{r_S}{r}(r^2 + a^2)}{2r^2} \implies \dot{r} = \pm \frac{\sqrt{\frac{r_S}{r}(r^2 + a^2)}}{r}, \quad (38.27)$$

where we made use of Eq. (38.26) and $\Delta = r^2 + a^2 - r_S r$. Next, Eq. (38.17) simplifies to,

$$\dot{\phi} = \frac{r_S a}{r \Delta}. \quad (38.28)$$

Combining Eqs. (38.27) and (38.28) one obtains,

$$\frac{d\phi}{dr} = \pm \frac{r_S a}{\Delta \sqrt{\frac{r_S}{r}(r^2 + a^2)}}. \quad (38.29)$$

which – when the minus sign is chosen – agrees with (38.9). Integrating (38.29) from $r = \infty$ to $r = r_0$ gives,

$$\Delta\phi(r_0) = a \int_{r_0}^{\infty} \frac{\sqrt{r_S r}}{(r - r_-)(r - r_+)\sqrt{r^2 + a^2}} dr. \quad (38.30)$$

where we wrote $\Delta = (r - r_-)(r - r_+)$, where

$$r_{\pm} = \frac{r_S}{2} \left[1 \pm \sqrt{1 - \left(\frac{2a}{r_S}\right)^2} \right], \quad (r_- < r_+, \quad r_S \geq 2a) \quad (38.31)$$

are the two event horizons of the Kerr black hole. From (38.30) we see that $\Delta\phi$ diverges logarithmically as $r_0 \rightarrow r_+$, the leading order term being,

$$\Delta\phi(r_0 \rightarrow r_+) \simeq a \frac{r_+}{r_S} \ln \left(\frac{r_0 - r_+}{r_+} \right), \quad (38.32)$$

where we made use of $\sqrt{r_+^2 + a^2} = \sqrt{r_S r_+}$. In order to find out whether the particle will fall into the black hole, we need to find out how $\phi = \phi(t)$. From (38.16) and (38.17) for the case at hand one gets,

$$\frac{d\phi}{dt} = \frac{\frac{r_S a}{r}}{r^2 + a^2 + \frac{r_S a^2}{r}}, \quad (38.33)$$

where $r = r(t)$. In order to integrate (38.33) we need to know $r = r(t)$, which can be obtained from Eqs. (38.16) and (38.27),

$$\frac{dr}{dt} = -\frac{\Delta \sqrt{r_S(r^2 + a^2)}}{r^{3/2} \left(r^2 + a^2 + \frac{r_S a^2}{r} \right)} \implies \Delta t(r_0) = \int_{\infty}^{r_0} \frac{r^{3/2} \left(r^2 + a^2 + \frac{r_S a^2}{r} \right) dr}{(r - r_-)(r - r_+) \sqrt{r_S(r^2 + a^2)}}. \quad (38.34)$$

Clearly, $\Delta t(r_0)$ diverges logarithmically as $r_0 \rightarrow r_+$, *i.e.* $\Delta t \sim \ln(r_0 - r_+)$, such that an asymptotic observer will never perceive a freely falling object to cross the event horizon. This conclusion is of course correct only if one neglects the (classical) backreaction of the falling object. When this backreaction is accounted for, its mass (or energy) will induce a slight increase in the event horizon r_+ , $\delta r_+ \simeq 2G_N m$, such that the test observer will – after a finite time $\Delta t \sim \ln(2G_N m/r_+)$ – be swallowed by the Kerr black hole. An analogous conclusion can be reached for $\Delta\phi$.

Notice that it is enough to know how long it will take to reach the outer event horizon $r = r_+$ to figure out whether an asymptotic observer will see an object falling into a Kerr black hole.

(f) To obtain the radii of circular orbits, one ought to find extrema of the effective potential (38.26), which (with $\Delta = r^2 + a^2 - r_S r$) can be also written as,

$$V_{\text{eff}} = -\frac{r_S}{2r} + \frac{1}{2} (\ell^2 - a^2(E^2 - 1)) \frac{1}{r^2} - \frac{1}{2} (\ell - aE)^2 \frac{r_S}{r^3}. \quad (38.35)$$

Taking a derivative,

$$V'_{\text{eff}} = \frac{\partial V_{\text{eff}}}{\partial r} = \frac{r_S}{2r^2} - (\ell^2 - a^2(E^2 - 1)) \frac{1}{r^3} + \frac{3}{2} (\ell - aE)^2 \frac{r_S}{r^4} = 0. \quad (38.36)$$

The condition $V'_{\text{eff}} = 0$ can be also written as the following (quadratic?) equation for r ,

$$\begin{aligned} r^2 - \frac{2}{r_S} [\ell^2 - a^2(E^2 - 1)] r + 3(\ell - aE)^2 &= 0 \\ \implies r_{\pm} &= \frac{1}{r_S} [\ell^2 - a^2(E^2 - 1)] \pm \sqrt{\frac{1}{r_S^2} [\ell^2 - a^2(E^2 - 1)]^2 - 3(\ell - aE)^2}. \end{aligned} \quad (38.37)$$

(g) For extremal black holes $a = r_S/2$, the two event horizons coincide, and are equal to $r_{\pm}^H = r_S/2$, and (38.37) reduces to,

$$r_{\pm} = \frac{1}{r_S} \left[\ell^2 - \frac{r_S^2}{4}(E^2 - 1) \right] \pm \sqrt{\frac{1}{r_S^2} \left[\ell^2 - \frac{r_S^2}{4}(E^2 - 1) \right]^2 - 3 \left(\ell - \frac{r_S}{2} E \right)^2}. \quad (38.38)$$

In the classical, nonrelativistic limit, $r_c = 2\ell^2/r_S = \ell^2/(G_N M) \gg r_S$, $|E - 1| \ll 1$, one obtains,

$$\begin{aligned}
r_{\pm} &= \frac{\ell^2}{r_S} \left[1 \pm \sqrt{1 - \frac{3r_S^2}{\ell^2} \left(1 - \frac{r_S}{2\ell}\right)^2} \right] = \frac{r_c}{2} \left[1 \pm \sqrt{1 - \frac{6r_S}{r_c} \left(1 - \frac{\sqrt{r_S}}{\sqrt{2r_c}}\right)^2} \right] \\
\Rightarrow r_+ &\simeq r_c \left[1 - \frac{3r_S}{2r_c} + \frac{3}{2} \left(\frac{2r_S}{r_c}\right)^{\frac{3}{2}} \right], \quad r_- \approx \frac{3}{2}r_S \left[1 - \sqrt{\frac{2r_S}{r_c}} + \frac{2r_S}{r_c} \right], \quad (38.39)
\end{aligned}$$

where we took $E = 1$. The r_+ solution corresponds to the stable circular orbit, which in the limit when $r_c/r_S \rightarrow \infty$ reduces to the classical circular orbit and r_- corresponds to the innermost unstable circular orbit, which in the same limit reduces to $r_- \rightarrow 3r_S/2 = 3G_N M$, which is also the radius of the unstable circular orbit of light. When compared to the equivalent orbits around a Schwarzschild black hole, the term $\propto \left(\frac{r_S}{r_c}\right)^{\frac{3}{2}}$ in r_+ and the term $\propto \sqrt{\frac{r_S}{r_c}}$ in r_- are absent in the case of the Schwarzschild black hole.